Terms

Time	Difficulty	Steps	You will need
10 min	Easy	10 terms	Nothing — just read

Ten words that carry the whole method. Everything else is built from these.

entropy
The randomness from the dice — the raw material the 24 words are built from.

airgap
Tails booted with networking **Disabled**. The computer forgets the session on shutdown.

2-of-3
Any two of three keys can spend. **One alone cannot.**

m/48'/0'/0'/2'
The wallet's path. **Confirmed on screen**, never typed from memory. Not a secret.

PSBT
The unsigned payment file, carried on the USB. The seed never leaves the offline computer.

descriptor / xpub
The wallet map / a watch key. Both **watch**, neither **spends**.

BIP39
The 2,048-word list the 24 words come from.

hash · fingerprint
The file's fingerprint must match the project's published one — **every character**.

signature
The project confirming the file. **"Good signature"** — or the file is deleted.

zero-trust
Nobody's word is taken. **Two tools must agree.**

← 02 · Before you begin | Step 3 of 15 — Checklist | 05 · Check the files →

Checklist

Time	Difficulty	Steps	You will need
30 min + shopping	Moderate	11 items	This sheet, A budget for the kit

Everything on the table before the first boot. Nothing is ordered mid-build.

On the tableThe dice table

- ■ **Five casino dice** with readable faces
- ■ **Pencil** and the recording grid
- ■ A **private room** — no camera

The computerOne machine, two boots

- ■ **Boot 1:** stays online, watch-only, never holds a seed
- ■ **Boot 2:** Tails USB, networking off
- ■ **Data USB:** carries only checked files

FilesChecked before anything goes offline

- ■ **Tails 7.13** · **Sparrow 2.5.5** (exact) · **Electrum 4.8.2**
- ■ **Ian Coleman** page (offline only) · **dice converter** (in this kit)

MetalThree plates, three places

- ■ **Three titanium plates** — one seed on each
- ■ **StampSeed** kit holds 1 (buy 2 more) · **Cryptotag** holds 2 (buy 1 more) — don’t mix methods
- ■ Three places, **far enough apart** that one disaster can’t take all three

Nothing is bought mid-build. If a plate or USB is missing when its sheet arrives, stop and get it first.

← 03 · Checklist | Step 4 of 15 — Check the files | 04 · Two boots →

Check the files

Time	Difficulty	Steps	You will need
30 min	Careful	7 sections	The downloaded files, Two verification tools

Every file gets checked before the computer goes offline. A file that fails is deleted. No exceptions — a compromised file compromises the funds.

What is required

- The downloads from the checklist: Tails 7.13, Sparrow 2.5.5, Electrum 4.8.2, the Ian Coleman page, and the dice converter from this kit.

Step 1 · Download only from the real sites.



NOT A LOOK-ALIKE PAGE · NOT FROM THIS SITE (EXCEPT THE DICE CONVERTER)

Sparrow must be 2.5.5 exactly. A newer download will not match — that is a stop, not an upgrade. The dice converter is the one file this site provides; its fingerprint is in VERIFY.md.

Step 2 · Compare the fingerprint. Every character.



compare with the fingerprint on the project's own site

EVERY CHARACTER MATCHES · ONE DIFFERS = DELETE

Windows: PowerShell → Get-FileHash. Mac: Terminal → shasum -a 256. Paste the result next to the fingerprint published on the project's own site.

Step 3 · Check the signature. Want: "Good signature".

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```
$ gpg --verify sparrow-2.5.5-manifest.txt.asc
gpg: Good signature
gpg: WARNING: This key is not certified with a trusted signature!
```

that warning is normal — the fingerprint match is what matters

Import the signing key, confirm its fingerprint, then verify. The "not certified" warning is normal on first import. What matters: the line says Good signature and the fingerprint matches.

Sparrow signing key fingerprint: D4D0D3202FC06849A257B38DE94618334C674B40. Electrum (ThomasV): 6694D8DE7BE8EE5631BED9502BD5824B7F9470E6.

Step 4 · Run the kit check.

```
$ ./verify_kit.sh
tails-amd64-7.13.img … PASS
sparrowwallet-2.5.5 … PASS
dice-converter … PASS
```

Put the downloads next to verify_kit.sh and run it. It checks the pinned fingerprints. One failure — the work stops.

Later · Two tools must agree. Every time.



SAME BITS SAME WORDS · SAME SEED SAME ADDRESSES · SAME PAYMENT DETAILS

Same dice bits → same 24 words in both tools. Same seed → same first addresses in Sparrow and Electrum. The wallet map shows those same addresses. Before signing, both tools show the same destination, amount, change, and fee. If they differ — stop.

STOP

Stop if

Next. Two boots. Flash Tails only after this page is done.

← 05 · Check the files | Step 5 of 15 — Two boots | 06 · Make the three keys →

Two boots

Time	Difficulty	Steps	You will need
45 min	Careful	4 sections	Everyday computer, Tails USB

Boot 1 · Network on

Watches balances. Builds a payment. Never holds a seed.

- Verify files
- Watch-only wallet
- Export unsigned file

then

shut down

Boot 2 · Networking off

This stick is the offline computer. It borrows the hardware, then forgets.

Networking off

- One seed in memory
- Sign the file
- Full shutdown after

One computer, used twice. The first boot watches the wallet and broadcasts finished payments — it never holds a seed. The second boot is the Tails USB: networking off, exactly one seed, then a full shutdown. The seed lives in memory only. Shutdown wipes it.



What "airgapped" means here

Yes:

- Networking Disabled on the Tails welcome screen, before the desktop.
- Files cross on the data USB — payment files, the map, checked programs.
- The online computer stays watch-only.

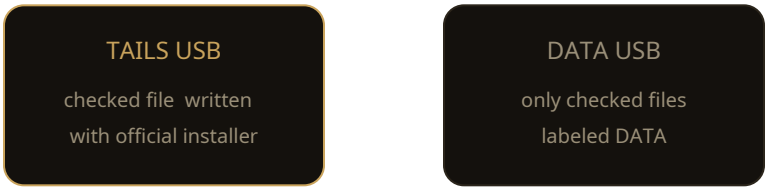
Not airgapped:

- Turning networking off after Sparrow opened.

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- Unplugging a disk that was online yesterday.
- A photo of a seed. A password manager, printer, or cloud note as backup.

Prepare the two USBs



NO PERSISTENT STORAGE — THE DRIVE STAYS AMNESIC

Download Tails only from the official site. Check the signature and checksum before writing. Write it with the official installer, label it TAILS. Do not create Persistent Storage — skip the passphrase. The data USB gets only files that already passed the checks.

Every offline session

1. Boot the Tails drive. Networking Disabled. Don't unlock Persistent Storage.
2. Copy only the file that session needs from the data USB.
3. Do the work: convert dice, restore one seed, sign one file, or export the map.
4. Copy the result back to the data USB. Shut down fully — dark screen before the drive comes out.

STOP

Stop if

Stronger (optional): a dedicated old laptop with the Wi-Fi card physically pulled. Ethernet never plugged. The Tails file is still checked, networking still Disabled, data USB still clean — the stick remains the only wire.

Next. The three keys are made next. A word is not written until Tails is running with networking turned off.

← 04 · Two boots | Step 6 of 15 — Make the three keys | 07 · Build the 2-of-3 →

Make the three keys

Time	Difficulty	Steps	You will need
30–45 min	Moderate	4 sections	5 dice, Paper and pen, The worksheet

11

10

01

01

11

10

10

01

256-BIT STRING

11 00 10 01 01 01 11 00 10 11 00 10 00 01 10
…keep rolling… **256 / 256 BITS**

/ 256

/ 256

/ 256

/ 256

Roll as many times as needed to get **256 bits** of entropy — about 32 shakes.

This work starts after Tails has booted with networking set to Disabled, and before any plate is stamped. Five casino dice are required, and every face has to be readable. Faces 1 through 4 are kept. Faces 5 and 6 are rolled again. The work is done three times, for three keys, and the computer is shut down fully between them. An established private setting is required, and there cannot be a camera in the room.

The faces

- 1 is 00. 2 is 01. 3 is 10. 4 is 11.
- 5 and 6 are not bits. Roll those dice again.
- Read the dice left to right, the same way every shake.
- Example: 4, 6, 1, 3, 2 becomes 11, skip the 6, then 00, 10, 01.
- Write only 0 and 1. Roll exactly 256 bits — 128 pairs, about 32 shakes — then stop. More bits do not add security (everything folds down to 256), and extra bits make the two tools below disagree, which stops the work for no reason.

Turn the bits into 24 words

scan for the full sheet

1. 01 · Boot Tails. Networking Disabled. Copy the checked converter and the Ian Coleman page from the data USB to the Desktop.
2. 02 · Run the converter. Paste the bits. It prints 24 words. Word 24 is the check. Do not invent it. Do not “fix” it.
3. 03 · Open the Ian Coleman page in the offline browser. Entropy type: Binary. 24 words. Paste the same bits. The 24 words must match. If they differ, stop. Do not pick the prettier list.
4. 04 · Write the 24 words on the grid. Read them back out loud once.
5. 05 · Shut Tails down. Wait until the screen is dark. Then start the next key. Seed 2 is a new set of rolls. Seed 3 is a new set of rolls. Never continue a list from the previous boot.

```
$ cd ~/Desktop$ python3 bip39_dice_converter.py
```

The last word is a check

The 24 words are not 24 free choices. The last word carries a short check so a copying mistake is caught before metal is stamped. At least two of these checks are used. If they disagree, the work stops.

- Two tools, same bits, same 24 words. Converter, then the offline Ian Coleman page. Do not pick the prettier list.
- Let the wallet refuse a bad list. Sparrow or Electrum, networking off, will open a valid list and reject a broken one. If it will not open, do not change a word to force it.
- The math, if it needs to be seen. Twenty-four words are 264 bits. The first 256 are the dice. The last 8 are the check. Only word 24 can change to make that true. It is not chosen. The converter appends it after the file has been checked.

Practice grid

Use the printed recording grid in the bench sheet PDF — each seed needs about 32 shakes for 256 bits. Count the pairs as you go and stop at 128.

Next. The 2-of-3 is built next. One seed is loaded per boot. Two lists are never loaded in one session.

← 06 · Make the three keys | Step 7 of 15 — Build the 2-of-3 | 10 · Rehearse on testnet →

Build the 2-of-3

Time	Difficulty	Steps	You will need
45–60 min	Careful	9 sections	The three paper keys, Verified Sparrow

Three public keys go in. No seed ever touches this build. What comes out is the wallet map — it can watch the money, never spend it.

You will boot Tails four times: once per seed to copy its public key, then once more with no seed to build the vault.

What is required

- The data USB with the checked Sparrow file (sparrowwallet-2.5.5-x86_64.tar.gz)
- The three paper seeds. Only one comes out per boot.

Step 1 · Boot Tails. Networking stays off.



IF THE NETWORK IS ON — STOP

The entire sheet happens with networking Disabled. Check it on every boot.

Step 2 · Open the terminal. Launch Sparrow.



Type these four lines, one at a time. Wait for the prompt to come back after each.

```
$ cd ~/Desktop

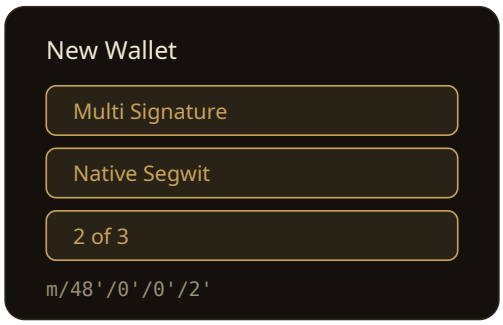
$ tar -xvf sparrowwallet-2.5.5-x86_64.tar.gz

$ cd Sparrow
```

```
$ ./bin/Sparrow
```

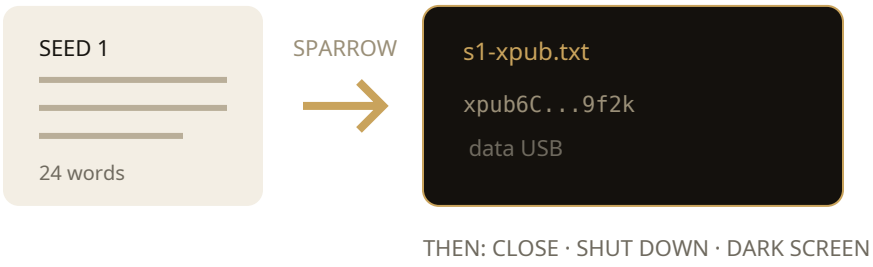
Wait 15–30 seconds. Sparrow opens. Type exactly — spaces count. If a line turns red, stop.

Step 3 · New wallet: 2 of 3, native Segwit.



File → New Wallet. Pick Multi Signature, Native Segwit, 2 of 3. The path must read m/48'/0'/0'/2'. Anything else — stop.

Step 4 · Keystore 1: Seed 1 in, public key out.



Keystore 1 → New or Imported Software Wallet → type Seed 1's 24 words. Export the xpub. Save it to the data USB as s1-xpub.txt. Close Sparrow. Shut down. Wait for a dark screen.

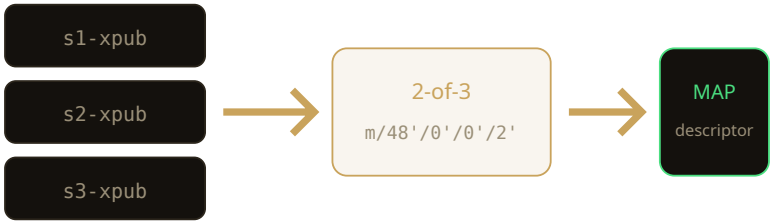
Step 5 · Twice more. One seed per boot.



Boot again for Seed 2 → s2-xpub.txt. Shut down. Boot again for Seed 3 → s3-xpub.txt. Shut down.

Step 6 · Fresh boot. No seed. Build the vault.

scan for the full sheet



NO SEED IS TYPED ON THIS BOOT

Boot Tails again — do not type any seed. New wallet → Multi Signature → Native Segwit → 2 of 3 → import the three xpub files. Confirm the path. Export the output descriptor to the data USB. That file is the wallet map.

Confirm the first address on a second screen.

Stop

Confirm the first address on a second screen.

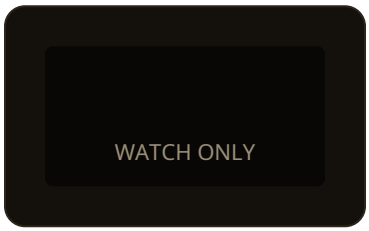
Sparrow built your wallet from your seeds — but if malware sat on the build machine, it could show you **an address it controls** while you trust it. A second screen the malware can't reach closes that hole. **Do not fund the wallet until this matches.**

1 · Build in Sparrow. Finish the 2-of-3 build on the airgapped machine as usual. Write down the **first receiving address** it shows — the full string, not the QR code.

2 · Cross-check in Electrum. Rebuild the same 2-of-3 in Electrum offline from the same seeds and compare the first address. This catches entry mistakes.

3 · NEW — Display it on a second device. Load the wallet as **watch-only** (xpubs / wallet descriptor only — never seeds) on your phone or a second computer. Compare the first receiving address **character for character**. If all three agree, the coordinator didn't lie. If anything differs, stop and rebuild.

Step 7 · Watch-only on the everyday computer.



THIS COMPUTER NEVER HOLDS A SEED

scan for the full sheet

On the everyday computer, open Sparrow and import the map as a watch-only wallet. It can see the money but can never spend it.

STOP

Stop if

Next. Practice spend. A small payment goes out and comes back before the plates are stamped.

← 10 · Rehearse on testnet | Step 9 of 15 — Practice spend | 09 · Metal and the map →

Practice spend

Time	Difficulty	Steps	You will need
45 min	Careful	5 sections	Two seeds, Tails USB, A small test amount

SHUTDOWN · NEW SESSION

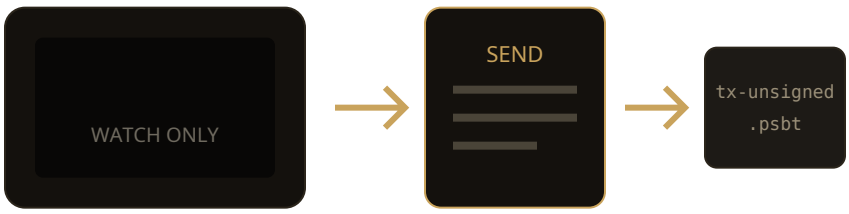
CREATE TRANSACTION

BROADCAST TRANSACTION

1. 1**Create.** Watch-only builds the transaction — the file is an empty outline.
2. 2**Seed 1 signs.** The file fills halfway. *1 of 2.*
3. 3**Shutdown.** The file stops at the red line — full shutdown, new session — then crosses deliberately.
4. 4**Seed 2 signs.** The file fills whole. *2 of 2.*
5. 5**Broadcast.** Up and right, across the airgap — the twice-signed file goes out.

A small amount — enough to pay a fee, not savings. Write the destination, amount, and fee on paper before anything is exported. Read those notes again before each signature.

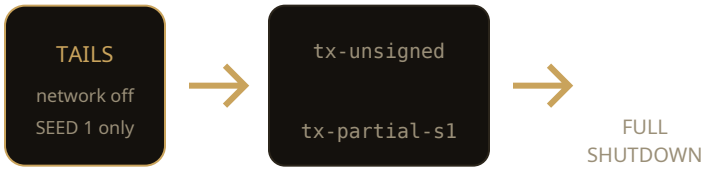
Step 1 · Build the spend — online, watch-only.



READ EVERY FIELD AGAINST THE PAPER NOTE

Open watch-only Sparrow. Send tab: destination, amount, fee. Finalize for signing. Export tx-unsigned.psbt to the data USB.

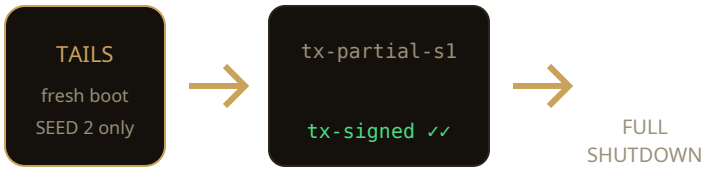
Step 2 · Seed 1 signs — Tails, networking off.



Boot Tails, networking disabled. Restore Seed 1 only. Open tx-unsigned.psbt, confirm the fields against the paper, sign. Export tx-partial-s1.psbt. Full shutdown.

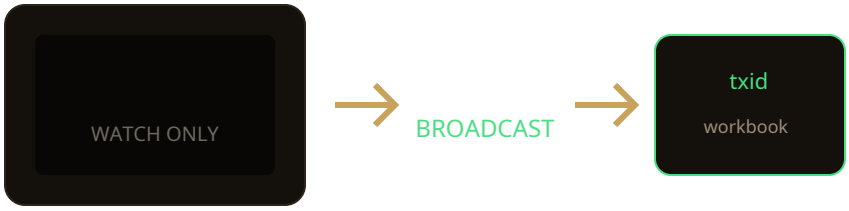
scan for the full sheet

Step 3 · Seed 2 signs — fresh boot.



Fresh Tails boot. Restore Seed 2 only. Open tx-partial-s1.psbt — confirm the first signature and that the spend is unchanged — sign. Export tx-signed.psbt. Full shutdown.

Step 4 · Broadcast — online computer.



Watch-only Sparrow: open the fully signed file, confirm the fields are unchanged, broadcast. Record the txid in the recovery workbook.

Keys on hardware later? Generate the seed on airgapped Tails first, then import the finished 24 words — never let the device roll its own entropy for this stack. Two keys on two brands (Foundation Passport, Blockstream Jade, BitBox02, SeedSigner/Krux — official shops only). Never three of the same box.

STOP

Stop if

Next. The plates are stamped next. A plate is not stamped until a small payment has gone out and come back.

← 08 · Practice spend | Step 10 of 15 — Metal and the map | 11 · Hide the wallet map →

Metal and the map

Time	Difficulty	Steps	You will need
1–2 hours	Moderate	4 sections	Titanium plates, Metal stamp set, The paper keys

S1

S2

S3

THREE PLATES THREE PLACES

A small payment has gone out and come back. Now the keys leave paper for good.

Three titanium plates. One seed per plate. Three separate places. If two plates sleep in the same house, the separation is gone — and the single point of failure is back.

The plates

- One seed per plate. Never two.
- StampSeed (recommended): the kit holds 1 plate, buy 2 more. Cryptotag: the kit holds 2, buy 1 more. Pick one method; don't mix them.
- Three places, far enough apart that one fire, flood, or burglary can't take all three.
- No photo, no note, no password manager, no email. Any copy outside the plates is a leak.

How to stamp

1. 01 · Private room, no camera. Plate on a solid block, each letter stamped deep.
2. 02 · Words numbered 1–24. Label the plate quietly — S1. No names, no "Bitcoin."
3. 03 · Read the plate back against the paper. A mis-stamp is a bricked backup.
4. 04 · Restore from the plate on offline Tails. First addresses must match.
5. 05 · Only then destroy the paper — and the bit notes with it.

The home plate

These 24 words, alone, open a small wallet. Leave a little bitcoin there: a thief who loads only those words may stop at what they find.

The same words plus a written passphrase open a modest spending wallet. The passphrase lives on paper — never on the plate, never in the same drawer. No passphrase on the 2-of-3 itself.

Where each plate lives

- **Plate A** — Seed 1 — home safe. Never Seed 2 or 3 here.
- **Plate B** — Seed 2 — off-site #1. Not the same address as A or C.
- **Plate C** — Seed 3 — off-site #2. Not the same address as A or B.

Print the output descriptor and keep it with Plate A, plus one copy off-site — never beside all three seeds. Hardware wallets don't replace these plates. Once a year: restore from the plates, receive dust, spend dust, update locations. Write a letter pointing to the packet and which two locations matter — not the words.

scan for the full sheet

Next. Hide the wallet map. The plates are not the map, and the map is not a seed.

← 09 · Metal and the map | Step 11 of 15 — Hide the wallet map | 12 · If I am gone →

Hide the wallet map

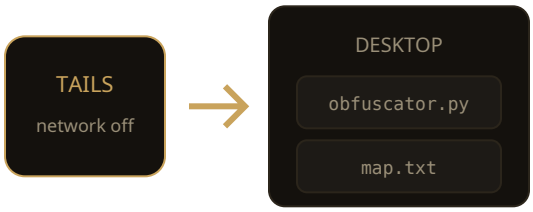
Time	Difficulty	Steps	You will need
30 min	Careful	7 sections	descriptor_obfuscator.py (verified), A dice passphrase

The map can watch the money. It cannot spend. Anyone who can read it can see the balance — so it never sits around as plain text.

What is required

- The wallet map on the data USB — called map.txt only while you work.
- descriptor_obfuscator.py from the kit. Checked before it's copied.
- A passphrase — not a seed, not the decoy passphrase, never stamped on a plate.
- Generate the passphrase with real dice: Diceware passphrase mapper — download the page and run it in an offline Tails session, never in an everyday browser.
- Tails, networking Disabled. Not the everyday computer.

Step 1 · Boot Tails. Copy the tool and the map.



NEVER OPEN THE MAP ON A PHONE

Boot Tails, networking Disabled. Copy the tool and map.txt to the Desktop.

Step 2 · Run the tool. Choose E. Enter the passphrase.

```
$ python3 descriptor_obfuscator.py
[E]ncrypt / [D]ecrypt? E
Paste wallet map: wsh(sortedmulti(2,…))
Passphrase: ••••••••••
wrote map.txt.gpg ✓
```

Open the terminal. Run the tool. Choose E. Paste the wallet map. Enter the passphrase — it is not a seed.

```
$ python3 descriptor_obfuscator.py
```

Step 3 · Copy the locked file. Delete the plain one.

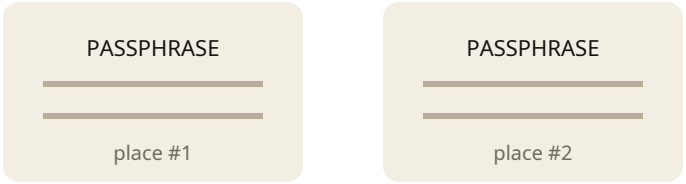
scan for the full sheet



EMPTY THE TRASH

Copy map.txt.gpg to two USB drives. Delete map.txt from the Desktop and the USB. Empty the trash.

Step 4 · Write the passphrase on paper. Twice.



AWAY FROM THE PLATES · AWAY FROM THE USBs

Two copies, two separate places. This paper is a single point of failure for every encrypted copy of the map — treat it like a plate.

Step 5 · Prove it opens. Check the first address.



On a later offline boot, choose D. The map must open, and the first address must match what you wrote down. If a plain copy was ever made — encrypt it again, then delete the plain one.

If the passphrase is lost, the coins aren't gone — as long as all three plates still exist. Three seeds rebuild the map exactly. Two seeds cannot. Write the path m/48'/0'/0'/2' on the inheritance sheet. It is not a secret.

STOP

Stop if

Next. If I am gone. That page is for a spouse. It does not contain words.

← 11 · Hide the wallet map | Step 12 of 15 — If I am gone | 13 · If you get stuck →

If I am gone

Time	Difficulty	Steps	You will need
1 hour	Moderate	6 sections	Your finished kit, Your trusted person

*This page is for someone who has **never used Bitcoin**. Read it before touching a plate. One plate cannot move the long-term fund — it needs **any two plates plus the wallet map**.*

Never type the words into a **phone** or a **website**.

Never **photograph** the plates.

Never put **both plates** into one device.

Never buy a device from a **marketplace** or a link in a message.



TWO PLATES + THE MAP — LEAVE THE THIRD PLATE WHERE IT IS

Find two plates in two places.

Leave the third where it is. You only need two.

Find the wallet map.

If it asks for a passphrase, that paper is stored **away from the plates**. Decrypt it on a trusted computer — or with someone who can follow the map sheet. The map is not the words.

Buy two signing devices — two different makers.

A **Jade** from Blockstream’s own shop. A **BitBox** from BitBox’s own shop. Not two of the same box.

One plate per device.

Plate A into one device. Plate B into the other. **Never both plates in one device.**

Load the map on both.

So both devices are looking at the same fund. The path is **m/48'/0'/0'/2'** — that line is not a secret.

Send a small test first.

If the test arrives, the fund can move. If the devices disagree, **stop** — don’t force it.

The home plate is a decoy.

Note

The home plate is a decoy. Its words alone open a small wallet — that is not the long-term fund. The fund still needs a second plate. Don’t mistake the home balance for the whole savings.

← 12 · If I am gone | Step 13 of 15 — If you get stuck | 14 · Tough questions →

If you get stuck

Time	Difficulty	Steps	You will need
10 min	Easy	7 questions	Nothing — find your question

Read the question that matches. When in doubt: ***stop is always a valid step.***

Q: What am I actually making?

Three keys, made at home with dice, on a computer with no internet. **Any two can spend; one cannot.** The keys are stamped into titanium and kept in three places. A hardware wallet is optional — and it never creates the keys.

Q: I already own a hardware wallet. Do I start there?

No. It stays in the drawer until the plates exist and a small payment has moved. If a key later goes on a device, the second key goes on a **different maker's** device. All three never go on hardware. **The plates are the backup.**

Q: The download is newer than the version on the sheet.

Stop. This kit checks Tails 7.13, Sparrow 2.5.5, Electrum 4.8.2. A newer file will not match — that is a stop, not an upgrade. Delete it.

Q: The two word lists don't match.

Stop. Don't pick the nicer list. Don't "fix" word 24. Roll again or find the copying mistake — a fixed word is a destroyed key.

Q: Can I keep the words in a password manager?

No. A photo, note, password manager, or email is one hacked account from the whole seed. **Words go on metal.** The map is encrypted; its passphrase is on paper, away from the plates.

Q: What if the map — or its passphrase — is lost?

Three plates rebuild the map exactly, so the coins aren't gone. But **two plates cannot** — the third key can't be derived from the other two. Back up the passphrase paper in two places, away from the plates. (Full rule on the map sheet.)

Q: What does my family do?

Hand them the **"If I am gone"** sheet. Two plates and the map. A Jade and a BitBox from the makers' own shops, one plate per device. No Tails. No typing words into a phone.

← 13 · If you get stuck | Step 14 of 15 — Tough questions | 15 · Recovery card →

Tough questions

Time	Difficulty	Steps	You will need
15 min	Easy	6 questions	Nothing — read and discuss

The questions people use to poke holes in the method. Only what is true.

Q: You typed the seed on a laptop. Isn’t that exposed?

Typed on Tails, networking off, one seed per boot, then a full shutdown — Tails forgets the session. A hardware keylogger is still possible. But one captured seed **still can’t spend**: the fund needs two. Never type two seeds on one boot.

Q: Why not start with a hardware wallet?

A signer can join later; it is not how these keys are made. The randomness is **dice you can watch** — no chip to trust, no firmware to audit.

Q: Is dice randomness weaker than a hardware RNG?

No. **256 bits is 256 bits**, whatever made them.

Q: If the map passphrase is lost, are the coins gone?

Not with all three plates — they rebuild the map exactly. But two plates plus a lost passphrase means the addresses are **unrecoverable**. Back up the passphrase paper in two places, away from the plates and USBs.

Q: What if someone finds one plate?

They can’t spend — one key alone opens nothing. But **never keep two plates in one place**: that’s the whole lock in one drawer.

Q: Shouldn’t I add a passphrase to the multisig?

No. The security is the **distribution**, not one more secret. A passphrase on the 2-of-3 is a single point of failure in disguise — lose it and two plates aren’t enough. The home plate’s small decoy wallet is the separate, optional place for a passphrase.

That is the whole method. Everything else is commentary.